

CHAINLAB

Blockchain Security and Scams

Threat models, red flags and defensive habits

FREE BLOCKCHAIN EDUCATION RESOURCE

This guide is designed for learning and revision. It explains foundational concepts and highlights practical limitations and safety considerations.

Learning objectives

- Identify frequent blockchain-related attack patterns.
- Use a basic threat-modeling process.
- Apply practical steps to reduce exposure.

1. Threat modeling

- Identify assets, actors, entry points, trust assumptions and potential impacts. Consider the wallet, device, frontend, contract, RPC provider, bridge, exchange and social accounts.
- Prioritize controls based on the value at risk and the likelihood and impact of plausible failures.

2. Common scams

- Phishing sites imitate legitimate services to steal credentials or induce signatures.
- Rug-pull patterns can involve misleading claims, concentrated control, blocked selling or liquidity removal. The term is broad; examine specific mechanics rather than relying on labels.
- Impersonation scams use fake support agents, giveaway claims, urgent warnings or copied social profiles.
- Malicious airdrops or tokens may direct users to unsafe sites or exploit confusing wallet interactions.

3. Technical risks

- Smart-contract bugs, compromised admin keys, oracle manipulation, bridge vulnerabilities, insecure dependencies and infrastructure outages can affect users.
- Audits, bug bounties, monitoring, least privilege and tested incident procedures can reduce risk but do not eliminate it.

4. Defensive habits

- Navigate through verified bookmarks or official channels. Never share a seed phrase. Avoid signing unexpected messages or transactions.
- Check addresses and permissions; revoke unnecessary approvals using trusted tools; keep devices and software updated; use separate wallets for experimentation when suitable.
- Treat guaranteed profits, pressure to act immediately and requests to pay a fee to unlock a prize as warning signs.

5. Incident response

- If a secret may be compromised, use a clean and trusted device to move remaining assets to a newly secured wallet when safe and technically appropriate.
- Disconnect suspicious applications, revoke permissions where possible, preserve transaction IDs and evidence, and contact legitimate platform support through verified channels.
- Be wary of recovery services demanding upfront payments or seed phrases.

Review questions

- What is a threat model?
- Give three social-engineering warning signs.
- Why can an audit not guarantee safety?

Educational notice: This material is for general education only. It is not financial, legal, tax or investment advice. Verify current network and project details before acting.